

NIRVANA – Master Specification

Volume 4: Financial Data Model & Database Schema

Purpose

Define the master data model that powers every recommendation. All calculations must use structured, normalized data instead of unstructured AI reasoning.

Core Database

Primary entities: Users, Families, Goals, Assets, Liabilities, Transactions, Holdings, MarketData, FinancialProducts, Recommendations, Reports, Advisors and AuditLogs.

Stocks Schema

Fields: ISIN, Symbol, Exchange, Sector, Industry, Market Cap, CMP, 52W High/Low, PE, PB, EPS, ROE, ROCE, Dividend Yield, Beta, Alpha, Sharpe, Volatility, Currency, Country, ESG Score, Analyst Rating, Historical Prices.

Mutual Fund Schema

Fields: AMC, Fund Name, Category, Benchmark, Fund Manager, NAV, AUM, Expense Ratio, Exit Load, Lock-in, Minimum SIP, Minimum Lumpsum, 1Y/3Y/5Y/10Y CAGR, Alpha, Beta, Sharpe, Sortino, Holdings, Sector Allocation, Riskometer.

ETF Schema

Fields: Underlying Index, NAV, Market Price, Tracking Error, AUM, Liquidity, Expense Ratio, Spread, Dividend Yield, Historical Returns.

Fixed Income

Tables for Bonds, Treasury Bills, G-Secs and FDs. Include issuer, rating, coupon, maturity, yield, duration, liquidity, tax treatment and lock-in.

Alternative Assets

Gold (ETF/SGB/Physical), Silver, REITs, InvITs, Real Estate and optional Crypto. Capture valuation, liquidity, taxation, appreciation assumptions and allocation limits.

Insurance

Life, Health, Critical Illness, Accident, Motor and Home Insurance. Fields include premium, cover, riders, exclusions, claim settlement ratio, waiting period and renewal data.

Loans

Home, Car, Personal, Education and Business Loans. Store lender, principal, outstanding, EMI, tenure, rate type, prepayment rules and tax benefits.

Credit Cards

Bank, Network, Joining Fee, Annual Fee, Waiver Rules, Cashback, Reward Rates, Lounge Access, Forex Markup, Milestone Benefits, Insurance, Eligibility, Income Requirement, Credit Score Requirement, Partner Offers and Validity.

Government Schemes

PPF, EPF, NPS, SCSS, SSY, NSC, RBI Bonds and other schemes with eligibility, lock-in, taxation, returns and contribution limits.

Macro Data

NIFTY, Sensex, VIX, Repo Rate, Inflation, GDP, Bond Yields, Gold Price, USD/INR and other economic indicators with timestamps.

Normalization Layer

All provider-specific fields from Investing.com, Paisabazaar and Excel must be mapped to a common internal schema before entering the recommendation engine.

Relationships

Users own portfolios; portfolios contain holdings; holdings reference products; products reference live market data; recommendations reference user profile, goals, risk score and market snapshot.

Refresh Strategy

Support manual and scheduled updates. Admin 'Update Data' refreshes live providers, validates records, removes duplicates, updates timestamps and falls back to cached Excel if synchronization fails.